

RESPONSE TO THE R18 REFEREE REPORT ON NEURAL BELLMAN OPERATORS

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1. OVERVIEW

We thank the referee for recognizing the positive high-dimensional menu evidence and for distinguishing a valid retrospective certificate from a prospective computation. This response addresses the advisory report of 5 October 2026 on the original Neural Bellman Operators paper. The report was commissioned by the repository owner, not the Econometric Society. The reviewed publication and report are pinned in the source baseline.

The revision develops centered decision loss, continuation transport and refresh, true-objective stopping, and a completed query-volume comparison. It retains every prior economic model and all favorable, unfavorable and unresolved scientific records. Four detailed components move intact to the supplement. The integrated computation replays the already frozen design and the same three streams; it is not counted as additional independent observations. All 252 service identities and failed checks are retained. The reported machine clocks belong to that particular complete execution.

B1. BEYOND AN IDENTICAL-FUTURE MENU

The main article now proves a continuation-transport theorem for bounded changes in future transitions and rewards. A second form treats recursive evaluation operators on a verified invariant utility domain. Together with the centered decision-loss proposition, these results define a nontrivial neighborhood of counterfactual economies in which an anchor continuation can be reused, and an explicit condition for refresh. This answers the theorem route requested in the report rather than inferring broad applicability from a finite menu. An explicit finite-feature corollary also derives sufficient fitting and Raw simulation budgets, accounts for dictionary

construction and legitimate cache reuse, and gives a complete-work threshold. Its assumptions are not asserted for an arbitrary all-layer neural fit. The original controlled-economy, recursive-utility, preference, temporal-self and game models remain. Their individual domains and deviation criteria are not waived. The new finite-support study is one capital application, not a fabricated experiment across all those models.

Locations: [3](#), [2](#), [S.28](#).

B2. SYMMETRIC RISK AND ITS CONNECTION TO DECISIONS

The new completed study evaluates every scalar predictor at the same exogenous lower, middle and upper actions, and separately at that predictor's own selected action. It reports absolute continuation risk, centered risk, own-action risk, finite-menu payoff loss and binary/menu misclassification. Full finite-law means provide the target, so a noisy squared simulation outcome is not mislabeled absolute risk. Raw caches are regenerated across three complete fitting/cache streams. The risk-to-decision proposition explains exactly which finite-menu loss is bounded by centered error; the primary scalar-interval guarantee is obtained independently from the true objective. Three fixed streams are retained individually and are not presented as precise inference over every future cache distribution. The earlier conditional comparisons remain unchanged, including the absence of a favorable NBO comparison at the 64-path budget.

Locations: [1](#), [XIV](#), [10](#).

B3. REAL STOPPING RATHER THAN A RETROSPECTIVE SELECTED CLOCK

Seven procedures now execute a frozen sequence of budgets. Each service stops at its first true-objective certificate below the common tolerance, charges all attempted fits, queries, failed checks and durable records, and otherwise remains unresolved. Every query volume receives an actual service execution, not a prefix allocation from a longer run. The 252 service identities and all stage records are published. A deterministic outward certificate makes the stopping guarantee simultaneous across the attempted stages without a statistical optional-stopping shortcut. The theorem also states the conditional-error requirement for a stochastic implementation. The

complete comparison bill remains separate from an individual deployment bill: we do not charge only the retrospectively cheapest method for a search that ran them all. This integrated execution is a replay of the already frozen source and seeds, not an additional independent experiment.

Locations: 4, S.47, S.29.

B4. QUERY-VOLUME AMORTIZATION FRONTIER

The executed volumes are 1, 4, 16, 64, 256 and 1,024 in both dimensions. Tables report the actual complete work needed to return the declared accuracy, retain unsuccessful services, and separate fixed/cache/fitting work, queries and all verification checks. The strict break-even formula states what follows when a fixed-plus-linear cost model is justified; no such shape is imposed on the measured costs. Cost bounds require upper-versus-lower separation before a guaranteed ranking can be inferred. The tables identify an observed discrete frontier on this machine, not an invented continuous breakpoint between tested volumes. Conventional procedures are retained when they are cheaper.

Locations: 11, XII, XIII, S.47.

B5. CONTRIBUTION-ISOLATING COMPARATORS AND LITERATURE

We add quadratic and radial-basis postdecision regressions using the same rollout value and derivative labels as the scalar NBO, a shared state-and-task actor, and full-support enumeration alongside cached SAA. These complement the inherited vector-costate, materialized-Raw and direct-policy comparisons. The literature discussion now distinguishes postdecision approximation, regression continuation methods, fitted value iteration, actor-critic methods, simulation reuse and valid sequential assessment. The center/contrast identity, strong-concavity residual inequality, difference-constraint closure and probability tools are identified as standard ingredients. The claimed contribution is their economic linkage and completed comparison, not a new principle of optimality or a universal superiority theorem for neural regression.

Locations: 10, 3.

B6. ECONOMIC MEANING OF THE DECISION MENU AND TOLERANCE

The new exercise computes current withdrawal responses to temporary resource charges while preserving the capital production law and fixed future policy. Its task distribution is stated before the numerical procedures and is not represented as an estimated empirical distribution. We prove that the true optimal withdrawal is nonincreasing in the charge, give the interior derivative, report zero-charge counterfactuals using the same fitted objects, and convert the welfare-loss allowance into an external log-utility grant. The primary machine guarantee remains the outward welfare-loss bound; the grant display is a descriptive binary64 conversion. This supplies a interpretable model counterfactual and a compensated-loss normalization without inventing an observed tax experiment or an equilibrium welfare conclusion.

Locations: [27](#), [28](#), [S.48](#).

B7. RELATION TO HIGH-DIMENSIONAL CONTINUOUS-TIME CONTROL

The main target table separates prediction risk, finite-menu loss, historical catalogue regret, the new continuous scalar-interval regret, and the full adapted-diffusion or equilibrium criteria. The new certificate is proved for the true finite-law scalar objective and does not by itself certify vector-action or adapted continuous-time optimality. The original continuous-time formulation, transfer bounds, exact-trace HJB comparison, utility selection and application proofs are all retained, including their mixed numerical findings. We strengthen the shared economic decision account without transferring a finite certificate to an unverified control class.

Locations: [I](#), [S.28.5](#), [11](#).

B8. ORGANIZATION AND THE RETAINED SCOPE

The revised opening states the economic decision, accuracy target, reuse condition and complete-work question before the general operator material. Four detailed capital and implementation components are moved intact from the main article to the technical supplement, with their labels, equations, proofs and numerical statements preserved. The complete application companion remains a current

part of the paper, connected by the transport theorem and the application-specific evaluation domains. We retain Neural Bellman Operators as the title because the revision develops that original object rather than replacing it with another topic. A preservation audit checks all inherited scientific source files; a reorganization map identifies every moved component. This is an argument-led reorganization, not an attempt to erase unresolved results or claim an editorial acceptance decision.

Locations: [2](#), [S.26](#), [11](#).

M1. STATE THE ECONOMIC QUESTION BEFORE THE GENERAL MACHINERY

The introduction, target table and decision theorem now precede the retained general controlled-economy specification. They identify the evaluated future policy, the current task distribution, the primary economic loss and complete work. The title and original substantive models remain, while the common decision question supplies the organizing argument.

Locations: [1](#), [2](#), [3](#).

M2. EXECUTE A FROZEN WORK-TO-TOLERANCE PROCEDURE

The published executable sequence fixes method order, both fitting stages, query volumes, three streams, the loss tolerance, first-success rule and failure treatment. Every one of the 252 services is executed, and the first-success and timing identities are independently checked from the records. The original freeze is linked to the recovered code and exact economic-input hashes. The new descriptive design file transparently reconstructs those unchanged choices; it is not claimed to reproduce the missing original design file byte for byte.

Locations: [4](#), [S.29](#).

M3. MAKE QUERY VOLUME EXPLICIT

Both main frontier tables report the six registered volumes. The supplement reports every method–volume–dimension cell and the complete timing decomposition, including current-only and unsuccessful actor outcomes. Shared task prefixes facilitate comparison without being described as independent new task draws.

Locations: [XII](#), [XIII](#), [S.49](#).

M4. USE COMMON-ACTION AND OWN-ACTION DIAGNOSTICS

The full-support diagnostic evaluates all scalar predictors on common exogenous action arguments and on their own selected arguments. It distinguishes levels, centered contrasts, finite-menu decision loss and actual scalar certificates. Cache-specific and stream-specific observations remain in the JSON record rather than being collapsed into a claim about an unmeasured population.

Locations: [XIV](#), [1](#).

M5. INCLUDE CONVENTIONAL SURROGATES AND A SHARED ACTOR

Quadratic and radial-basis regressions, a shared task-conditioned actor and structure-exploiting enumeration are implemented, executed and retained with the neural and cached-simulation procedures. No comparator is discarded for reaching the target more cheaply. Fitting information and cache reuse rights are described in the main design.

Locations: [10](#), [XI](#).

M6. TRANSLATE THE TASK EXERCISE INTO ECONOMIC CONSEQUENCES

The task is a temporary-charge withdrawal counterfactual in the inherited capital economy. The true comparative-statics sign, approximate withdrawal responses and explicit external-compensation conversion make the accuracy target economically interpretable. The design does not purport to estimate tax incidence from data or a new equilibrium response of uncomputed agents.

Locations: [28](#), [S.48](#).

M7. SEPARATE ESTABLISHED TOOLS FROM THE NEW LINKAGE

The expanded discussion credits established postdecision, fitted-value, regression and actor–critic methods. The new theorem chain is presented as an explicit economic error and transport account using standard inequalities, rather than claiming novelty for centering, shortest paths, residual optimization or confidence sequences. Bibliographic details are checked against primary publisher or author sources.

Locations: [3](#).

M8. MAKE TARGET HIERARCHY VISIBLE

The main article contains a dedicated target table before the theory, and every new results table specifies its target and comparison class. True scalar-interval regret is the new primary target; continuation risk and exogenous finite-menu loss are secondary diagnostics. Historical catalogue and continuous-time accounts retain their separate notation and provenance.

Locations: [I](#), [XI](#), [XIV](#).

M9. RETAIN THE UNFAVORABLE FINDINGS

The original cheaper Raw/vector clocks, SAA's Long-design advantages, the one-cell least-cost-certified NBO finding, the lack of a 64-path NBO risk win, and cheaper unresolved candidates remain in the current manuscript. The new conventional-surrogate cost advantages and failed shared-actor checks are reported as well. No unsuccessful fit or verification stage is relabeled a successful replacement.

Locations: [1](#), [S.49](#).

M10. REDUCE REPETITION WITHOUT DISCARDING SCIENTIFIC MATERIAL

Four detailed components are relocated verbatim to a clearly indexed supplement section, while the main article places one decision-loss/transport/stopping chain and the primary comparisons first. The complete applications remain accessible in their current companion. The publication contains full proofs and all records rather than a shorter but scientifically incomplete replacement. The remaining judgment about breadth is left to the referee; file preservation and clean compilation do not themselves establish journal suitability.

Locations: [S.26](#), [3](#).